Sampling Methods and Representativeness

Advanced Topics: Methodology, Philosophy, and Research Design

1. Population vs. Sample

Population: All individuals/objects about which a statement is to be made.

Sample: Subset of the population studied.

Representativeness: Sample mirrors the population in relevant characteristics.

Sampling error: Deviation of sample statistic from true population value (by chance).

Systematic error (bias): Deviation due to distorted selection (not reduced by chance).

2. Probability Sampling

Simple random sample: Every element has equal, known selection probability. Gold standard.

Stratified random sample: Population divided into strata, random selection per stratum.

Cluster sample: Random selection of groups, then all within group.

Multi-stage sample: Combination of cluster and random (e.g., school -> class -> student).

3. Non-Probability Sampling

Convenience sample: What is readily available. Highest bias risk.

Quota sample: Gender, age by population quotas. No random selection within quotas.

Snowball sample: Interviewees refer to others. For hard-to-reach populations.

Purposive sample: Experts deliberately selected. Typical for qualitative research.

4. Weighting

Design weighting: Corrects different selection probabilities.

Post-stratification: Adjustment to known population distributions (raking).

Caution: Over-weighting individual cases increases sampling error.